46000 Introduction to Wind Energy
Report 2
Aerodynamics of wind turbine rotors

Figure 1: Materials for a 2-bladed rotor.

The report deals with the design, manufactoring and test of a 2-bladed rotor for a wind turbine. A piece of wood with the following characteristics is supplied: length $L = 0.5\ m$, thickness $t = 0.02\ m$, max width $W = 0.04\ m$. The task is to manufacture a rotor that can deliver the largest possible power. The rotor is tested in an open-jet wind tunnel at a wind speed of 3.2 m/s.

The theory

The rotor can be designed by optimum rotor theory. The essential formulas can be summarised as follows:

$$16a^3 - 24a^2 + a(9 - 3x^2) - 1 + x^2 = 0 \tag{1}$$

$$a' = \frac{1 - 3a}{4a - 1} \tag{2}$$

$$\tan\phi = \frac{1 - a}{(1 + a')x} \tag{3}$$

$$c = \frac{8\pi r a \sin^2\phi F}{B(1 - a)\left[C_l \cos\phi + C_d \sin\phi\right]} \tag{4}$$

where

$$x = \lambda \frac{r}{R} \tag{5}$$

$$\lambda = \omega R / V_0 \tag{6}$$

$$F = \frac{2}{\pi} \arccos\left(\exp(-f)\right) \tag{7}$$

$$f = \frac{B}{2} \frac{R - r}{r \sin\phi} \tag{8}$$

The optimum design of a rotor is a function of the tip speed ratio λ. Figure 2 shows C_P as function of λ for different values of C_l/C_d.

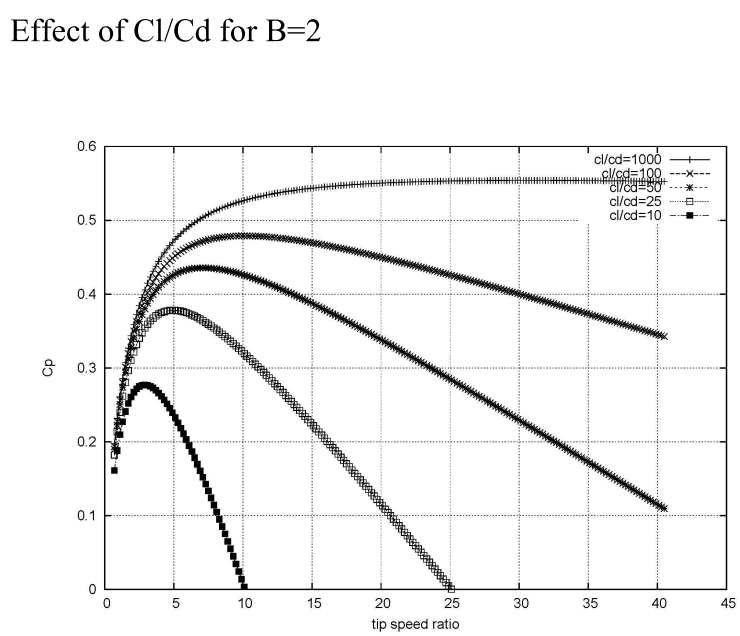

Figure 2: Approximate relation for C_P versus λ for two-bladed optimum rotors.

The tasks

Part 1: The velocity triangle

1. Look up the SD7003 with 8.5% thickness airfoil shape on the Airfoil Tools website. Plot its shape. Mark the leading edge, the trailing edge, the chord line and the camber line.

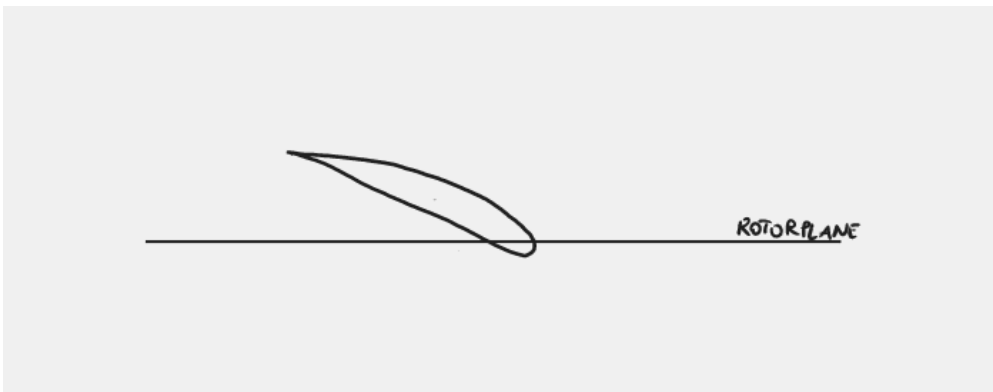

Figure 3: Sketch of wind turbine airfoil.

2. Draw a sketch like the one in Figure 3. Draw the velocity triangle that consists of the incoming wind speed, the rotationally induced wind speed and the relative wind speed, at the blade tip, assuming a tip speed ratio of $\lambda = 4$ and an axial induction of $a = 0.3$. The drawing must be at an aspect ratio of 1:1 (true angles). Mark the flow angle ϕ.

3. For question 2, calculate the required twist angle β to achieve an angle of attack of $\alpha = 4°$. Show your working. Mark the angle of attack α and the twist angle β on your drawing from question 2.

4. Plot the polar curves for C_l and C_d of the SD7003 airfoil from the *sd7003.csv* file at $Re = 30000$ between an angle of attack of $-5°$ and $15°$. Identify the stall region.

5. Make a copy of the sketch from question 2 and add the following vectors: lift force L, drag force D, resulting force and tangential force P_t. The relative size of the arrows should be consistent with an angle of attack of $\alpha = 4°$ at $Re = 30000$. Briefly describe each vector and explain the mutual relationship between them.

6. Make a stick 3D-model of a 2-bladed wind turbine with the materials supplied in the lecture. Make a copy of the drawing from question 5. Insert it in one of the blades of your model turbine so that the rotor turns clockwise when observed from upstream. Take a picture from the side of the rotor, and include it in the report.

Part 2: The rotor design

Point 7 to 9 concern the design of the rotor. They are best completed in an iterative manner, to achieve a suitable compromise between performance (C_p) and available chord length from the piece of wood.

7. By the aid of figure 2 and of the airfoil polar curves, choose a target value of C_l/C_d and next an operating tip speed ratio. Next compute the Reynolds number based on an approximate chord length $Re_c = cV_{rel}/\nu$. Argue for your choice of tip speed ratio.

8. Plot C_l/C_d as function of angle of attack. Plot also C_l versus angle of attack. Identify the point of operation. Argue for the decision of operation point.

9. As function of radius along the blade, r, compute

   - the induction factor a,
   - the rotational induction factor a',
   - the flow angle ϕ and the blade twist angle β
   - the chord length c
   - the local chord-based Reynolds number $Re_c = cV_{rel}/\nu$

   Present the results in four plots and discuss your design.

10. Pick three locations along the radius. Draw the velocity triangle at these three locations for the chosen tip speed ratio. The two axes must have same scale such that all angles are true.

11. Carve out your rotor in wood. Fabricate a couple of templates in paper in scale 1:1 and with equally scaled axes to help get the right shape. Document how you transferred the above design to the wood.

12. Measure the rotational speed when the rotor is exposed to wind. Discuss the performance of your rotor: was the rotor able to achieve the design tip speed ratio? How did the produced power relate to the design power? What are the reasons for differences between your design and achieved performance?